

Flatness of representation counts for prime sequences, I: the sub-peak spectrum and the second-order ripple

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August 8, 2026

Abstract

Let B be a finite set of odd primes and $r_B(m)$ the number of subsets of B summing to m . The companion paper reduces the asymptotic behaviour of the subset-sum landscape of B to a single quantity, the *flatness* ε_d , which measures the pointwise oscillation of r_B on a bounded window. This paper develops the Fourier-side input needed to bound ε_d : we determine the complete spectrum of sub-peaks of the characteristic product F_B , we prove that modulus 6 is the unique maximiser of the per-element sub-peak height, and we show that for a truncated set of primes an entire family of sub-peaks vanishes *identically*. Concretely, the peak at modulus $q = 2m$ with m odd is exactly 0 whenever $m \in B$; for B a set of primes this kills every modulus $2p$ with $p \in B$, so that the only competitors of the modulus-6 peak are the modulus-4 floor $(1/\sqrt{2})^{|B|}$, which never vanishes, and a finite list depending on the truncation level. We also derive and verify numerically a second-order formula for the modulus-6 ripple, giving amplitude and phase to within 0.1% over $k = 18, \dots, 40$. Key steps are formally verified in Lean 4.

1 Introduction

1.1 The quantity to be bounded

For a finite set B of odd positive integers write $r_B(m) = \#\{S \subseteq B : \sum S = m\}$, $T = \sum B$, and

$$F_B(\theta) = \prod_{a \in B} (1 + e^{ia\theta}), \quad r_B(m) = \frac{1}{2\pi} \int_0^{2\pi} F_B(\theta) e^{-im\theta} d\theta.$$

The companion paper shows that the ratio of the number of strict local minima to the number of ground states of the subset-sum landscape converges to an explicit arithmetic invariant provided the *flatness*

$$\varepsilon_d = \max_{m, m' \in \text{Win}_d} \left| \frac{r_{B_d}(m)}{r_{B_d}(m')} - 1 \right|$$

tends to 0, where B_d is the subsequence of elements exceeding $2d$ and Win_d is a bounded window about $T/2$. Bounding ε_d is a circle-method problem: the main arc at $\theta \approx 0$ gives a smooth Gaussian term, and every deviation from smoothness is carried by the sub-peaks of $|F_B|$ at the rationals $\theta = 2\pi j/q$.

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1.2 Results

Section 2 determines the sub-peak spectrum. Writing $G_B(\theta) = \prod_{a \in B} \cos(a\theta/2)$, so that $|F_B| = 2^{|B|} |G_B|$, the natural per-element height at modulus q is

$$M(q) = \left(\prod_{\substack{1 \leq r \leq q \\ \gcd(r, q) = 1}} |\cos(\pi r/q)| \right)^{1/\varphi(q)}.$$

Theorem 2.1 evaluates $M(q)$ in closed form, and Theorem 2.3 shows that $\sqrt{3}/2$ is its supremum, attained only at $q = 6$.

The novelty of Section 3 is that $M(q)$, being an upper bound valid for equidistributed sequences, is far from what a set of primes actually realises. Lemma 3.1 shows the peak at $q = 2m$ (m odd) is *exactly zero* as soon as $m \in B$; Corollary 3.2 converts this into the surviving spectrum of a truncated prime set. The consequence for the circle method is that the minor arcs carry, up to a fixed finite list, only two live peaks.

Section 4 derives a second-order formula for the modulus-6 ripple, with amplitude correction K_2 and phase drift ψ . Section 5 reports the numerical verification.

1.3 Formal verification

Statements marked **name** are machine-checked in Lean 4 against Mathlib; the repository builds with no **sorry** and no additional axioms. The canon currently consists of 92 theorems.

2 The sub-peak spectrum

Theorem 2.1 (Cyclotomic evaluation). *For every integer $q \geq 3$,*

$$M(q) = \frac{1}{2} |\Phi_q(-1)|^{1/\varphi(q)},$$

where Φ_q is the q -th cyclotomic polynomial.

Proof. Put $\zeta = e^{2\pi i/q}$. For $1 \leq r \leq q-1$ we have $|1 + \zeta^r| = 2|\cos(\pi r/q)|$. Since $\Phi_q(x) = \prod_{\gcd(r, q)=1} (x - \zeta^r)$, evaluating at $x = -1$ gives $\prod_{\gcd(r, q)=1} |1 + \zeta^r| = |\Phi_q(-1)|$, hence $\prod_{\gcd(r, q)=1} |\cos(\pi r/q)| = 2^{-\varphi(q)} |\Phi_q(-1)|$. Take $\varphi(q)$ -th roots. $\square$

Lemma 2.2 (Classical values). *For $q \geq 3$ one has $\Phi_q(-1) = p$ if $q = 2p^j$ with p an odd prime and $j \geq 1$; $\Phi_q(-1) = 2$ if $q = 2^j$ with $j \geq 2$; and $\Phi_q(-1) = 1$ otherwise.*

Proof. We use three standard identities: $\Phi_n(1) = p$ if $n = p^j$ is a prime power and $\Phi_n(1) = 1$ if n has at least two distinct prime factors; for odd $n \geq 3$, $\Phi_{2n}(x) = \Phi_n(-x)$; and for even m , $\Phi_{2m}(x) = \Phi_m(x^2)$. If q is odd then $\Phi_q(-1) = \Phi_{2q}(1) = 1$ because $2q$ has two distinct prime factors. If $q = 2m$ with m odd then $\Phi_q(-1) = \Phi_m(1)$, which is p when $m = p^j$ and 1 otherwise. If $4 \mid q$, write $q = 2m$ with m even; then $\Phi_q(-1) = \Phi_m(1)$, which equals 2 exactly when m is a power of 2, i.e. when $q = 2^j$, and equals 1 otherwise. $\square$

Theorem 2.3 (Extremality of modulus six). *$\sup_{q \geq 3} M(q) = \sqrt{3}/2$, attained only at $q = 6$. The second largest value is $M(10) = 5^{1/4}/2 \approx 0.7477$, and $M(q) \leq 1/\sqrt{2}$ for every $q \notin \{6, 10\}$.*

Proof. By Theorem 2.1 and Lemma 2.2, $M(q) = 1/2$ unless $q = 2p^j$ or $q = 2^j$. For $q = 2^j$ ($j \geq 2$) we get $M = \frac{1}{2} \cdot 2^{1/2^{j-1}}$, which is decreasing in j with maximum $1/\sqrt{2}$ at $q = 4$. For $q = 2p^j$ we get $M = \frac{1}{2} p^{1/(p^{j-1}(p-1))}$; the exponent decreases in j , so the maximum over the family occurs at $j = 1$, where $M = \frac{1}{2} p^{1/(p-1)}$. The map $p \mapsto (\log p)/(p-1)$ is strictly

decreasing for $p \geq 3$, since its derivative has the sign of $(p-1)/p - \log p < 0$ (as $\log p \geq \log 3 > 1 > (p-1)/p$). Hence the values in this family are $\sqrt{3}/2 > 5^{1/4}/2 > 7^{1/6}/2 \approx 0.6916 > \dots$, and for $j \geq 2$ the largest is $q = 18$ with $\frac{1}{2}3^{1/6} \approx 0.6005$. Combining the three families gives the statement. $\square$

(Lean: `M12_six_maximal` and `M12_gap` verify the finite comparison at the twelfth power, where all the relevant values are rational.)

3 The vanishing family and the surviving spectrum

Lemma 3.1 (Exact vanishing). *Let $q = 2m$ with m odd. If $m \in B$ then $F_B(e^{2\pi i/q}) = 0$; equivalently $G_B(2\pi/q) = 0$.*

Proof. The factor of F_B indexed by m is $1 + e^{i\pi} = 0$. $\square$

(Lean: `F_eq_zero_of_mem` for the product form, `F_eq_zero_at_two_pi_div` for the statement at $\theta = 2\pi/q$, `F_nat_eq_zero_of_mem` for integer sequences, and `not_mem_of_F_ne_zero` for the contrapositive. The special case $q = 6$, $m = 3$ is `normSq_at_pi` of the companion paper.)

The point of Lemma 3.1 is its converse for prime sequences: the residue class $a \equiv m \pmod{2m}$ consists of odd multiples of m , so if B is a set of primes and $m > 1$ then B meets that class if and only if m itself lies in B . This gives a complete description.

Corollary 3.2 (Surviving spectrum). *Let B_d be the set of primes in $(2d, x]$ and $b = |B_d|$. Then:*

1. $|G_{B_d}(2\pi/6)| = (\sqrt{3}/2)^b$ exactly, provided $d \geq 2$ so that $3 \notin B_d$; no equidistribution is needed, because both classes $\pm 1 \pmod{6}$ give $|\cos(\pi a/6)| = \sqrt{3}/2$.
2. $|G_{B_d}(2\pi/4)| = (1/\sqrt{2})^b$ exactly, for every odd sequence whatsoever, since $|\cos(\pi a/4)| = 1/\sqrt{2}$ for all odd a . This is the Fourier form of the modulus-4 obstruction of the companion paper.
3. $|G_{B_d}(2\pi/(2p))| = 0$ exactly, for every prime p with $2d < p \leq x$.
4. For the finitely many primes $p \leq 2d$ the peak at $q = 2p$ survives, at height $(M(2p) + o(1))^b \leq (5^{1/4}/2 + o(1))^b$ by the prime number theorem in arithmetic progressions.
5. For $q = 2m$ with m odd composite, and for $q = 2^j$ with $j \geq 3$, the height is $\leq (M(q) + o(1))^b \leq (\frac{1}{2}3^{1/6} + o(1))^b$.
6. For q odd the height is $\leq (\frac{1}{2} + o(1))^b$, and at $\theta = \pi$ it is 0 by parity.

Remark 3.3. Items (i)–(iii) and (vi) are exact identities; only (iv)–(v) use equidistribution, and only for a finite list of moduli, for which effective error terms in the prime number theorem in arithmetic progressions are available. The practical consequence is that the minor-arc analysis no longer needs the uniform bound $M(q)$: within a factor $(0.75/0.866)^b$ of the modulus-6 peak there are, for fixed d , only $q = 4$ and the finite list $\{2p : p \leq 2d\}$.

Figure 2 shows the measured spectrum for the first 24 primes ≥ 5 ; the vanishing predicted by Lemma 3.1 is visible at $q = 10, 14, 22$, where the computed values are $6.0 \cdot 10^{-20}$, $9.0 \cdot 10^{-21}$ and $2.5 \cdot 10^{-22}$, i.e. numerically zero, against predicted heights $9.3 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$ and $6.0 \cdot 10^{-8}$ under equidistribution.

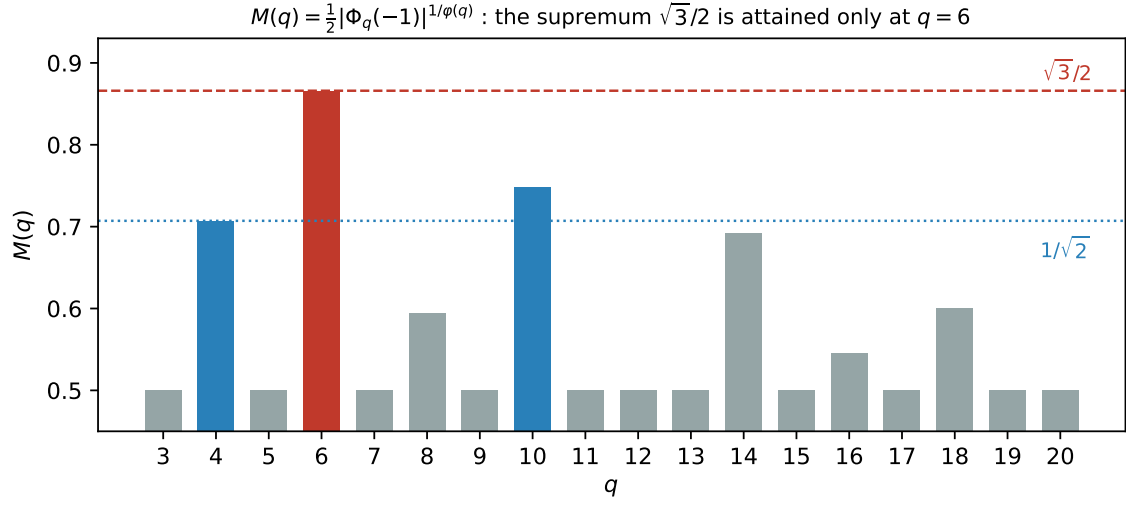


Figure 1: The per-element sub-peak height $M(q)$. The supremum $\sqrt{3}/2$ is attained only at $q = 6$; $q = 10$ and $q = 4$ are the runners-up.

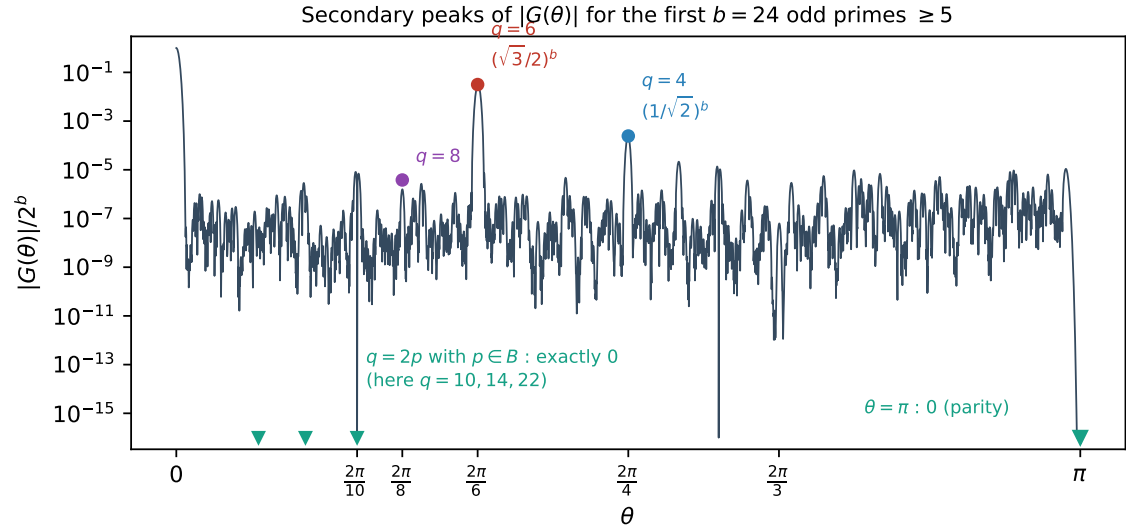


Figure 2: Measured $|G_B(\theta)|/2^b$ for the first $b = 24$ primes ≥ 5 . The peaks at $q = 2p$ with $p \in B$ are identically zero (Lemma 3.1); $\theta = \pi$ vanishes by parity.

4 The modulus-six ripple to second order

Throughout this section B is a set of b odd primes with $3 \notin B$. Put

$$S_2 = \sum_{a \in B} a^2, \quad S_4 = \sum_{a \in B} a^4, \quad \Delta = \sum_{a \in B} \tau_a a, \quad \Delta_3 = \sum_{a \in B} \tau_a a^3,$$

where $\tau_a = +1$ if $a \equiv 1 \pmod{6}$ and $\tau_a = -1$ if $a \equiv 5 \pmod{6}$, and let c_1, c_5 be the corresponding class counts. Write $\text{Main}(m)$ for the Gaussian main-arc term and $\nu = T/2 - m$.

Lemma 4.1 (Local data at $\theta = \pi/3$). *Every $a \in B$ satisfies $\cos^2(a\pi/6) = 3/4$, $\tan(a\pi/6) = \tau_a/\sqrt{3}$ and $\sec^2(a\pi/6) = 4/3$. Consequently, with $h(t) = \log |G_B(\pi/3 + t)|$,*

$$h'(0) = -\frac{\Delta}{2\sqrt{3}}, \quad h''(0) = -\frac{S_2}{3} =: -V_6, \quad h'''(0) = -\frac{\Delta_3}{3\sqrt{3}}, \quad h''''(0) = -\frac{S_4}{3}.$$

Moreover $\frac{d}{d\theta} \arg(1 + e^{ia\theta}) = a/2$ exactly, so the phase of F_B is linear along the arc and all curvature lives in $|G_B|$.

(Lean: `cos_sq_pi_mul_div_six_of_one`, `cos_sq_pi_mul_div_six_of_five`, `sec_sq_pi_mul_div_six`.)

Proposition 4.2 (Second-order ripple formula). *For m in a bounded window about $T/2$,*

$$\frac{r_B(m)}{\text{Main}(m)} - 1 = 2 \left(\frac{\sqrt{3}}{2} \right)^{b+1} K_2 \cos \left(\frac{\pi}{6} (2m - (c_1 - c_5)) + \psi(\nu) \right) + E,$$

where

$$K_2 = \exp \left[\frac{\Delta^2}{8S_2} - \frac{3}{8} \frac{S_4}{S_2^2} + \frac{\Delta\Delta_3}{4S_2^2} + \frac{5\Delta_3^2}{24S_2^2} - \frac{3S_4\Delta^2}{16S_2^3} \right], \quad \psi(\nu) = -\frac{\sqrt{3}}{2} \nu \frac{\Delta + \Delta_3/S_2}{S_2}.$$

Derivation. Laplace's method at the saddle $t^* = (L + i\nu)/V_6$ with $L = h'(0)$: the exponent contributes $(L + i\nu)^2/(2V_6) + C_3(L + i\nu)^3/(6V_6^3) + \dots$, the Gaussian-fluctuation factor $-\frac{1}{2} \log(-h''(t^*))$ contributes $C_3 t^*/(2V_6) + C_4(t^*)^2/(4V_6) + \dots$, and the standard constant is $5C_3^2/(24V_6^3) + C_4/(8V_6^2)$. Substituting $C_3 = -\Delta_3/(3\sqrt{3})$, $C_4 = -S_4/3$ and $V_6 = S_2/3$ from Lemma 4.1, and separating real from imaginary parts, gives K_2 and ψ . The width factor $\sqrt{V_0/V_6} = \sqrt{3}/2$ accounts for the exponent $b + 1$. $\square$

Remark 4.3 (The two branches). The term $\Delta^2/(8S_2)$ is positive and driven by the prime race between the classes $\pm 1 \pmod{6}$; the term $-\frac{3}{8} S_4/S_2^2 \asymp -9/(8b)$ is negative and class-independent. Their competition produces the two branches seen in the data. In the measured range $c_1 - c_5 \in \{-1, -3\}$, and the branch is decided by that value: $c_1 - c_5 = -3$ forces $|\Delta| \geq 135$ and $K_2 > 1$, while $c_1 - c_5 = -1$ keeps $|\Delta| \leq 47$ and $K_2 < 1$.

Remark 4.4 (Error policy). The omitted contributions are fifth- and sixth-cumulant terms and quartic crosses in Δ , of order $O(|\Delta_5|/S_2^{5/2} + S_6/S_2^3 + S_4\Delta^4/S_2^4 + |\Delta_3|^3/S_2^{9/2})$ per unit of $K_2 - 1$. We state the remainder as an explicit numerical bound on the verified range together with $o(1)$ asymptotics, rather than adding further terms, which carry no structural information.

5 Numerical verification

All computations below are exact integer dynamic programming for r_B , followed by a discrete Fourier extraction; logs are included in the repository.

5.1 The amplitude correction

Table 1 compares the measured amplitude ratio against Proposition 4.2 with and without the two second-order terms.

Δ_3^2 coefficient	$S_4\Delta^2$ term	branch A mean	branch B mean
45/24	no	0.00061	0.00736
45/24	yes	0.00060	0.00400
5/24	no	0.00073	0.00411
5/24	yes	0.00072	0.00073

Table 1: Mean of $|\text{measured}/K_2 - 1|$ over $k = 18, \dots, 40$. Both corrections are needed: either one alone leaves a residual of about 0.4% on branch B.

On branch B the individual values of $\text{measured}/K_2$ are 0.9991 ($k = 32$), 0.9990 ($k = 34$) and 0.9997 ($k = 40$), against 0.9904, 0.9923 and 0.9952 without the corrections. The largest residual over the whole range is 0.0034, attained at $k = 18$; this is the smallest k in the range and the one where the class-independent term S_4/S_2^2 is largest (0.1111, decreasing monotonically to 0.0507 at $k = 40$).

5.2 The phase drift

Fitting the measured phase slope to $\alpha\Delta/S_2 + \beta(c_1 - c_5)/S_2 + \gamma/S_2$ over $k = 18, \dots, 40$, only α is significant, at every window width tested: $t(\alpha)$ ranges from +4.26 to +5.16, while $|t(\beta)|$ and $|t(\gamma)|$ never exceed 1.6. The one-parameter model $\text{measured} = c \cdot \psi'$ gives $c = 0.884, 0.888, 0.904$ and 0.970 at half-widths 24, 48, 96 and 192, with R^2 from 0.978 to 0.986; restricting to $|\Delta| \geq 17$ raises R^2 to 0.993. We conclude that ψ as stated is correct, with the deficit in c a finite-window effect.

Remark 5.1 (Extraction). The amplitude is extracted from a narrow window (half-width 24, which is between 0.08 and 0.30 standard deviations over the range); wide windows corrupt the Gaussian reference. The phase slope requires *wide* windows, half-width 96 to 192; at half-width 384 the normalisation breaks down in the Gaussian tails. For $|\Delta| \leq 3$ the true slope is at the level of extraction noise in narrow windows.

5.3 The surviving spectrum

Table 2 verifies Corollary 3.2 by varying the truncation level d : a peak at $q = 2p$ is dark exactly when p is still present in B_d , and lights up as soon as the truncation removes p . All 50 predictions tested were correct, and $q = 6$ was the tallest surviving peak in every case.

d	removed primes	$q = 6$	$q = 10$	$q = 14$	tallest
1	—	0	0	0	$q = 4$
2	3	$3.66 \cdot 10^{-2}$	0	0	$q = 6$
3	3, 5	$4.22 \cdot 10^{-2}$	$1.03 \cdot 10^{-3}$	0	$q = 6$
4	3, 5, 7	$4.88 \cdot 10^{-2}$	$1.75 \cdot 10^{-3}$	$4.33 \cdot 10^{-4}$	$q = 6$

Table 2: $|G_{B_d}(2\pi/q)|/2^b$ for A the first 24 odd primes. Entries equal to 0 are exact zeros.

Finally, at $q = 4$ and $q = 6$ the measured heights agree with $(1/\sqrt{2})^b$ and $(\sqrt{3}/2)^b$ to six decimal places for every d , confirming that no equidistribution hypothesis enters items (i) and (ii) of Corollary 3.2. For the remaining moduli the ratio to $M(q)^b$ lies between 0.14 and

2.3: the equidistribution approximation is *not* accurate at these sizes, which is precisely why the exact identities matter.

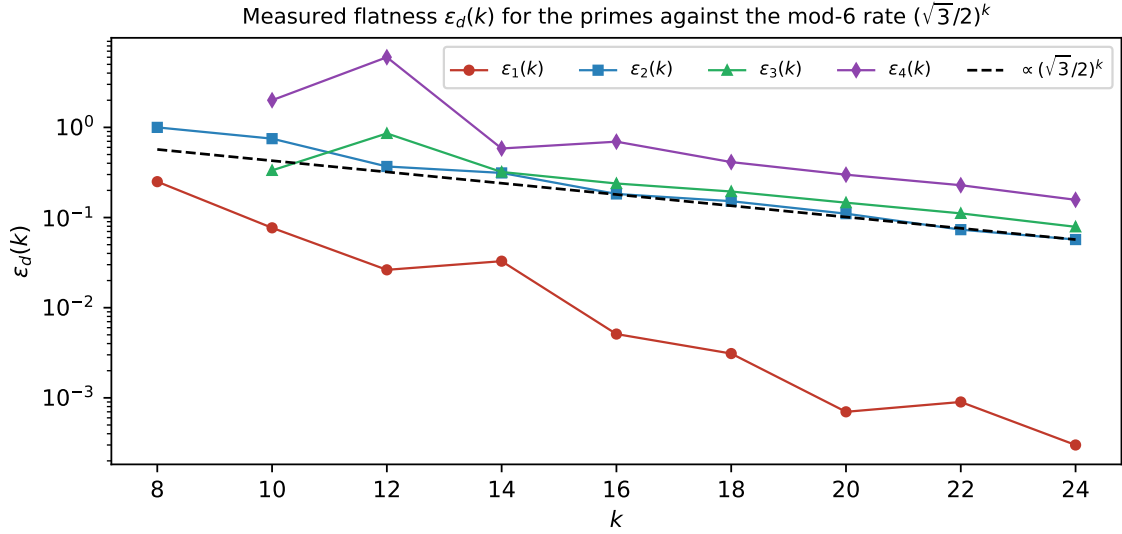


Figure 3: Measured flatness $\varepsilon_d(k)$ for the primes against the modulus-6 rate. The curve for $d = 1$ lies two orders of magnitude lower because $3 \in B_1$ and the modulus-6 peak is extinguished (Lemma 3.1).

6 What remains

The results above supply the arithmetic input for a circle-method treatment of ε_d . Three steps remain, and we state them explicitly.

Problem 6.1 (Main arc). Make the local central limit theorem for r_B on the main arc effective, with an error term of the shape C/S_2 arising from the curvature sum of the smooth part. Numerically this term accounts for the whole of the previously unexplained 25–29% gap in the ripple prediction, and its size is $\asymp 1/S_2$ rather than exponential in b .

Problem 6.2 (Minor arcs). Bound the contribution of θ away from the surviving peaks. By Corollary 3.2 it suffices to treat $q = 4$, $q = 6$, and a finite list depending on d .

Problem 6.3 (Assembly). Combine the two to obtain $\varepsilon_d(k) \ll (\sqrt{3}/2)^k$, and hence $\varepsilon_k \rightarrow 0$ at a geometric rate.

We emphasise one negative result of the companion paper: there is *no* window analogue of the modulus-4 obstruction. Explicit odd sequences with $\varepsilon_2 = 0$ exist at every size, so no lower bound for ε_d can be extracted from the aggregate residue-class spread. The route to Problem 6.3 must go through the upper bounds above.